

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12410788
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Li; Ran et al.

Plunger pump and pump station

Abstract

A plunger pump includes a pump head assembly, which includes: a pump head body, a chamber in the pump head body; a liquid suction valve assembly, a liquid discharge valve assembly, and a separation sleeve for separating the liquid suction valve assembly from the liquid discharge valve assembly that are located in the chamber. The chamber includes: a liquid suction valve base mounting portion for mounting the liquid suction valve assembly, a separation sleeve mounting portion for mounting the separation sleeve, a liquid discharge valve base mounting portion for mounting the liquid discharge valve assembly, and a liquid suction chamber on a lower side of the liquid suction valve base mounting portion and in communication with an emulsion supply device by a pipeline.

Inventors:	Li; Ran (Beijing, CN), Wei; Wenshu (Beijing, CN), Ye; Jian (Beijing, CN), Liu; Hao (Beijing, CN), Wu; Mengyu (Beijing, CN), Chen; Rongming (Beijing, CN), Wang; Dalong (Beijing, CN), Chen; Jingbin (Beijing, CN)
Applicant:	CCTEG BEIJING TIANMA INTELLIGENT CONTROL TECHNOLOGY CO., LTD. (Beijing, CN); BEIJING CCRI-TIANMA AUTOMATION TECHNOLOGY CO., LTD. (Beijing, CN)
Family ID:	72657474
Assignee:	CCTEG BEIJING TIANMA INTELLIGENT CONTROL TECHNOLOGY CO., LTD. (Beijing, CN); BEIJING CCRI-TIANMA AUTOMATION TECHNOLOGY CO., LTD. (Beijing, CN)
Appl. No.:	18/006332
Filed (or PCT Filed):	April 12, 2021
PCT No.:	PCT/CN2021/086360
PCT Pub. No.:	WO2022/016911

PCT Pub. Date: January 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230313792 A1	Oct. 05, 2023

Foreign Application Priority Data

CN	202010718136.1	Jul. 23, 2020
----	----------------	---------------

Publication Classification

Int. Cl.: F04B53/10 (20060101); F04B9/04 (20060101)

U.S. Cl.:

CPC F04B53/1022 (20130101); F04B9/045 (20130101); F04B53/1087 (20130101); F04B53/109 (20130101);

Field of Classification Search

CPC: F04B (9/045); F04B (15/02); F04B (53/10-1097)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1507448	12/1923	Aldrich	251/360	F16K 17/04
2146709	12/1938	Bird	92/128	F04B 53/164
2559659	12/1950	Redman	417/568	F04B 53/1025
2766701	12/1955	Giraudeau	277/525	F04B 53/14
3053500	12/1961	Atkinson	251/332	F16K 1/38
4188174	12/1979	Perkins	137/246.22	F04B 53/1087
4520837	12/1984	Cole	417/568	E21B 33/13
6910871	12/2004	Blume	417/454	F04B 53/007
7364412	12/2007	Kugelev	417/454	F04B 53/1032
8147227	12/2011	Blume	137/512	F04B 53/164
10094376	12/2017	Vicars	N/A	F04B 39/122
2017/0051732	12/2016	Hemandez	N/A	F04B 17/06
2017/0159655	12/2016	Morreale	N/A	F16K 15/026
2018/0355449	12/2017	Underys	N/A	C21D 9/00
2019/0055937	12/2018	Parmeshwar	N/A	F04B 9/045
2020/0362678	12/2019	Lesko	N/A	F04B 9/045
2021/0239229	12/2020	Kelton	N/A	F04B 1/0461

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2771507	12/2005	CN	N/A
2898367	12/2006	CN	N/A
205605419	12/2015	CN	N/A
205605419	12/2015	CN	N/A
206439170	12/2016	CN	N/A
206439170	12/2016	CN	N/A
109538460	12/2018	CN	N/A
209278105	12/2018	CN	N/A
111255678	12/2019	CN	N/A
111255678	12/2019	CN	N/A
211009053	12/2019	CN	N/A
2484949	12/2011	EP	N/A
3146210	12/2019	EP	N/A
WO-2014186512	12/2013	WO	N/A

OTHER PUBLICATIONS

WIPO, International Search Report for PCT/CN2021/086360, Jul. 16, 2021. cited by applicant
CNIPA, First Office Action for CN Application No. 202010718136.1, Dec. 10, 2024. cited by applicant

Primary Examiner: Comley; Alexander B

Attorney, Agent or Firm: The Sun IP Law

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. national stage entry of International Application No. PCT/CN2021/086360, filed Apr. 12, 2021, which claims priority to Chinese Patent Application No. 202010718136.1, filed Jul. 23, 2020, the entire disclosures of which are incorporated herein by reference.

FIELD

(2) The present disclosure relates to an emulsion pump station, and more particularly to a five-plunger emulsion pump and a pump station.

BACKGROUND

(3) Emulsion pump stations are essential and important equipment for fully mechanized coal mining faces, which provide hydraulic power for hydraulic supports of the faces and are the core of the hydraulic system for the whole faces. In recent years, with the increasing number of working faces with large mining height in China, higher requirements have been put forward for the reliability of emulsion pump stations, in order to meet the requirements of high support resistance and high working resistance of hydraulic supports with large mining height, as well as the requirements of fast moving and safe support.

(4) At present, the emulsion pump generally adopts large-flow emulsion pumps with a five-plunger structure or a seven-plunger structure, which have the advantages of high transmission efficiency, stable support, small axial force component of gear pair, and compact structure, and are widely

used.

(5) In the related art, a liquid suction valve core assembly and a liquid discharge valve assembly in a pump head body of a plunger pump are assembled from both sides of the pump head body, so during replacement of a liquid suction valve base, a liquid suction valve core, a liquid discharge valve base, and a liquid discharge valve core, parts such as a suction pipe, a suction box and the like need to be removed, which leads to poor operability, long maintenance time and high maintenance cost. Additionally, in the plunger pump of the related art, a reset spring mounting seat for the liquid suction valve core is generally designed at a lower part of the liquid discharge valve core. On one hand, the mass of the liquid discharge valve core is increased, and the inertia force becomes large; on the other hand, high-pressure liquid flow generates impact on the liquid suction valve core and the liquid discharge valve core, which are subjected to large force on one side, and when the valve core rises up or down, guide parts are subjected to eccentric wear, and in turn sealing annular surfaces undergo eccentric wear, which affects the service life. The pump head body and the suction box of the existing plunger pump are two independent parts, and during their installation, it is necessary to provide a long installation hole in the suction box, and the two parts are assembled or disassembled by a long screw, which leads to higher manufacturing and assembly costs.

(6) Moreover, current monitoring techniques for pump stations mainly include monitoring of gear oil in a crankcase (oil temperature monitoring, oil pressure monitoring, oil level monitoring), monitoring of water pressure at a hydraulic end, and motor temperature monitoring. Some patents also involve monitoring of vibration and oil quantity of mechanical equipment. However, due to the limitation of installation space and the influence of transmission signals, sensors are generally mounted outside the pump stations. At present, there is no monitoring of built-in precision core components such as suction and liquid discharge valve cores at the hydraulic end of the pump station.

SUMMARY

(7) A plunger pump includes: a crankcase assembly configured to be coupled to a main drive motor to transmit power; a pump head assembly configured to pump emulsion; and a hydraulic conversion assembly coupled between the crankcase assembly and the pump head assembly and configured to convert mechanical power of the crankcase assembly into a hydraulic change of the pump head assembly. The pump head assembly includes: a pump head having a chamber; a liquid suction valve assembly, a liquid discharge valve assembly and a separation sleeve separating the liquid suction valve assembly from the liquid discharge valve assembly, all of which are in the chamber. The chamber includes a liquid suction valve base mounting portion for mounting the liquid suction valve assembly, a separation sleeve mounting portion for mounting the separation sleeve, a liquid discharge valve base mounting portion for mounting the liquid discharge valve assembly, and a liquid suction chamber at a lower side of the liquid suction valve base mounting portion. The liquid suction chamber is in connection with an emulsion supply device through a pipeline. The liquid discharge valve base mounting portion is transitioned to the separation sleeve mounting portion through a stepped surface; a bore diameter of the liquid discharge valve base mounting portion is larger than a bore diameter of the separation sleeve mounting portion; and the bore diameter of the separation sleeve mounting portion is larger than or equal to a bore diameter of the liquid suction valve base mounting portion.

(8) The present disclosure also provides a pump station that includes a base; and a plunger pump and a main drive motor being both on the base, in which the main drive motor is configured to drive the plunger pump. The plunger pump includes: a crankcase assembly configured to be coupled to a main drive motor to transmit power; a pump head assembly configured to pump emulsion; and a hydraulic conversion assembly coupled between the crankcase assembly and the pump head assembly and configured to convert mechanical power of the crankcase assembly into a hydraulic change of the pump head assembly. The pump head assembly includes: a pump head

having a chamber; a liquid suction valve assembly, a liquid discharge valve assembly and a separation sleeve separating the liquid suction valve assembly from the liquid discharge valve assembly, all of which are in the chamber. The chamber includes a liquid suction valve base mounting portion for mounting the liquid suction valve assembly, a separation sleeve mounting portion for mounting the separation sleeve, a liquid discharge valve base mounting portion for mounting the liquid discharge valve assembly, and a liquid suction chamber at a lower side of the liquid suction valve base mounting portion. The liquid suction chamber is in connection with an emulsion supply device through a pipeline. The liquid discharge valve base mounting portion is transitioned to the separation sleeve mounting portion through a stepped surface; a bore diameter of the liquid discharge valve base mounting portion is larger than a bore diameter of the separation sleeve mounting portion; and the bore diameter of the separation sleeve mounting portion is larger than or equal to a bore diameter of the liquid suction valve base mounting portion.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) Embodiments of the present disclosure will be described in detail below with reference to the accompanying drawings, which will help to understand the objectives and advantages of the present disclosure, in which:
- (2) FIG. 1 is a sectional view of a specific embodiment of a plunger pump according to the present disclosure;
- (3) FIG. 2 is a structural diagram of a specific embodiment of a plunger pump according to the present disclosure;
- (4) FIG. 3 is a sectional view of a plunger head assembly of a plunger pump according to the present disclosure;
- (5) FIG. 4 is a partial schematic view illustrating a connection relationship between a plunger and a slider in a plunger pump according to the present disclosure;
- (6) FIG. 5 is a structural diagram of a liquid suction valve base in a plunger pump according to the present disclosure;
- (7) FIG. 6 is a structural diagram of a liquid suction valve core in a plunger pump according to the present disclosure;
- (8) FIG. 7 is a structural diagram of a liquid discharge valve base in a plunger pump according to the present disclosure;
- (9) FIG. 8 is a structural diagram of a liquid discharge valve core in a plunger pump according to the present disclosure;
- (10) FIG. 9 is a structural diagram of a liquid discharge valve core stop member in a plunger pump according to the present disclosure;
- (11) FIG. 10 is a structural diagram of a specific embodiment of a pump station according to the present disclosure.

DETAILED DESCRIPTION

(12) Technical solutions of the present disclosure will be clearly and completely described below with reference to the drawings. Obviously, the described embodiments are merely some rather than all of the embodiments of the present disclosure. Based on the embodiments of the present disclosure, all other embodiments obtained by those skilled in the art without creative effort fall into the scope of the present disclosure. In addition, technical features involved in different embodiments of the present disclosure described below can be combined with each other as long as they do not contradict with each other.

(13) The first technical problem to be solved by the present disclosure is that an unreasonable structure of the existing plunger pump leads to time-consuming and laborious disassembly and

assembly of the pump head.

(14) The present disclosure provides the following technical solutions aiming at the above technical problem.

(15) FIG. 1 and FIG. 2 show a specific embodiment of a plunger pump according to the present disclosure, and the plunger pump in this embodiment is a five-plunger emulsion pump. The plunger pump A includes three parts. A first part is a crankcase assembly **300** configured to be coupled to a main drive motor to transmit power; a second part is a pump head assembly **100** configured to pump emulsion; a third part is a hydraulic conversion assembly **200** configured to convert mechanical power of the crankcase assembly **300** into a hydraulic change of the pump head assembly **100**. One end of the hydraulic conversion assembly **200** is coupled to the crankcase assembly **300**, and the other end of the hydraulic conversion assembly **200** is coupled to the pump head assembly **100**.

(16) The structures of the above three parts and their connection will be described below.

(17) <Pump Head Assembly **100**>

(18) As shown in FIG. 3, the pump head assembly **100** includes: a pump head body **101**; a liquid suction valve assembly **102**, a liquid discharge valve assembly **103**, a separation sleeve **104** configured to separate the liquid suction valve assembly **102** from the liquid discharge valve assembly **103**, the liquid suction valve assembly **102**, the liquid discharge valve assembly **103** and the separation sleeve **104** being all on an inner side of the pump head body **101**; a liquid suction chamber blocking plate **112** configured to block a lower opening of the pump head body **101**; and a liquid discharge chamber blocking cover **105** configured to block an upper opening of the pump head body **101**.

(19) The pump head body **101** is a chamber structure with upper and lower openings. The chamber includes: a liquid suction valve base mounting portion **1012** for mounting the liquid suction valve assembly **102**; a separation sleeve mounting portion **1013** for mounting the separation sleeve **104**; a liquid discharge valve base mounting portion **1014** for mounting the liquid discharge valve assembly **103**; and a liquid suction chamber **1011** at a lower side of the liquid suction valve assembly **102**, in which the liquid suction chamber **1011** is in connection with an emulsion supply device (not shown) through a pipeline.

(20) In order to facilitate the installation of various components, the transition between the liquid discharge valve base mounting portion **1014** and the separation sleeve mounting portion **1013** is achieved by a stepped surface; a bore diameter of the liquid discharge valve base mounting portion **1014** is larger than a bore diameter of the separation sleeve mounting portion **1013**; and the bore diameter of the separation sleeve mounting portion **1013** is larger than or equal to a bore diameter of the liquid suction valve base mounting portion **1012**. In this way, when the liquid suction valve assembly **102**, the liquid discharge valve assembly **103** and the separation sleeve **104** are mounted, the liquid suction valve assembly **102** can easily enter the liquid suction valve base mounting portion **1012** through the liquid discharge valve base mounting portion **1014** and the separation sleeve mounting portion **1013**; the separation sleeve **104** enters the separation sleeve mounting portion **1013** after passing through liquid suction valve base mounting portion **1012**; and then the liquid discharge valve assembly **103** is mounted. As a result, the liquid suction valve assembly **102**, the liquid discharge valve assembly **103**, and the separation sleeve **104** can be assembled and disassembled through the upper opening of the chamber.

(21) Since the liquid suction valve assembly **102**, the liquid discharge valve assembly **103** and the separation sleeve **104** are all mounted from one side, components can be removed simply by opening an upper end of the pump head body **101** when the components in the pump head need to be repaired. Meanwhile, since the liquid suction valve assembly **102**, the liquid discharge valve assembly **103** and the separation sleeve **104** are all mounted from one side, axial fit relationship of various components can be easily guaranteed, and mutual interference caused by poor position matching in case of installation from both sides can be avoided.

(22) Meanwhile, the liquid suction chamber is formed directly inside the pump head body **101**, which can avoid problems of high manufacturing cost and time-consuming and laborious assembly due to detachable connection between a liquid suction part and the pump head body **101**.

(23) In addition, in the present disclosure, the liquid suction valve assembly **102** and the liquid discharge valve assembly **103** are separated by the separation sleeve **104** and are structurally independent, which can prevent liquid flow from impacting the liquid suction valve core and the liquid discharge valve core in the process of liquid suction and liquid discharge, especially in the process of high-pressure liquid discharge. Moreover, there is no unilateral stress problem, so that no eccentric wear occurs to guide parts during opening or closing of the valve core, and in turn sealing annular surfaces is not subjected to eccentric wear and have a longer service life.

(24) Specifically, the liquid suction chamber **1011** is formed as a spherical chamber structure, and a diameter of the spherical chamber larger than diameters of other parts of the chamber, to ensure that sufficient emulsion enters the chamber on an upper side of the liquid suction valve core during suction.

(25) The compositions and structures of the liquid suction valve assembly **102**, the liquid discharge valve assembly **103**, and the separation sleeve **104** will be described below.

(26) <Liquid Suction Valve Assembly **102**>

(27) As shown in FIG. 3, the liquid suction valve assembly **102** includes a liquid suction valve base **1021**, a liquid suction valve core **1022**, a reset spring mounting seat **1023** at the lower side of the liquid suction valve base **1021** and a first reset spring **1024**.

(28) The liquid suction valve core **1022** slides up and down along the liquid suction valve base **1021** by the action of hydraulic force, the first reset spring **1024** is used to provide downward elastic force to the liquid suction valve core **1022** and block the opening of the liquid suction valve base **1021**. The reset spring mounting seat **1023** is installed at the lower side of the liquid suction valve base **1021**. Compared with the related art, the lower end of liquid discharge valve core is designed with a spring mounting seat for resetting the liquid suction valve core **1022**. When the liquid suction valve core **1022** is reset, it is not affected by the action of the liquid discharge valve core, the liquid suction valve core has reliable valve closing, less impact and longer service life.

(29) Specifically, the reset spring mounting seat **1023** is fitted over the liquid suction valve core **1022**, both of the reset spring mounting seat **1023** and the liquid suction valve core **1022** can fitted by screw or interference fit. The reset spring mounting seat **1023** is installed on the liquid suction valve core **1022** to make the liquid suction valve assembly **102** completely independent of the liquid discharge valve assembly **103**, which is convenient for assembly and disassembly.

(30) Specifically, the liquid suction valve base **1021** is in clearance fit with the pump head body **101**, and the sealing connection is realized by a sealing ring to facilitate the liquid suction valve base **1021**. Specifically, the sealing connection style between the liquid suction valve base **1021** and the mounting portion **1012** of the liquid suction valve base, as a preference, there is a sealing groove on the outer side of the liquid suction valve base **1021**, and an O-shaped sealing ring and two sealing retaining rings are arranged in the sealing groove, and the two of the sealing retaining rings are arranged at the upper and lower sides of the O-shaped sealing ring respectively. Due to the high liquid pressure of emulsion pump, sealing rings made of polyoxymethylene are designed on both sides of the O-shaped sealing retaining ring to prevent from being damaged by the high-pressure emulsion.

(31) Specifically, as shown in FIG. 5, the liquid suction valve base **1021** includes: an outer sleeve **1021-1**, an inner sleeve **1021-2** and a transition arm **1021-3** connecting the inner sleeve **1021-2** and the outer sleeve **1021-1**.

(32) The outer wall of the outer sleeve **1021-1** is hermetically coupled to the mounting portion **1012** of the liquid suction valve base **1021**, the terminal of the inner wall of the outer sleeve **1021-1** is provided with a first matching surface **1021-4** for matching with the liquid suction valve core **1022**; the inner wall of the inner sleeve **1021-2** is slidably coupled to the liquid suction valve core

1022.

(33) As shown in FIG. 6, the liquid suction valve core includes: a valve core head **1022-1** and a valve core rod **1022-2**, the valve core head **1022-1** is configured for matching with outer sleeve **1021-1**, the valve core rod **1022-2** is slidably coupled to the inner sleeve **1021-2** of the liquid suction valve base **1021**.

(34) The liquid suction valve base **1021** mentioned above adopts the structure of inner side sleeved the outer side, and the liquid suction valve core **1022** adopts the structure of valve core head **1022-1** and valve core rod **1022-2**, so that the liquid suction valve core **1022** can reliably slide along the inner sleeve of the liquid suction valve base **1021**, thus preventing the liquid suction valve core from shaking radially relative to the liquid suction valve base in the related art.

(35) The valve core head **1022-1** is provided with a second matching surface **1022-3**, which matches with the first matching surface **1021-4**. The first matching surface **1021-4** and the second matching surface **1022-3** are formed as conical surfaces; the liquid suction valve core **1022** and the liquid suction valve base **1021** are matched by the conical surfaces, so that the sealing performance between them is better; meanwhile, the liquid suction valve core **1022** can be compensated automatically after being worn, and the service life is prolonged.

(36) In order to detect the impact force of the liquid suction valve core **1022**, as shown in FIG. 3, the bottom of the liquid suction valve base **1021** is provided with a pressure sensor **107**, the pressure sensor **107** is configured for detecting the impact force acting on the liquid suction valve base **1021**. The pressure sensor **107** monitors the impact force of each liquid suction and discharge valve core action in real time, and can also calculate the impact frequency according to the impact force variation curve. When the impact force and the frequency are greatly deviated from the normal state, it can be determined that the group of liquid suction and discharge valve cores have failed, and alarm is given to the operator to disassemble and check. The signal line of pressure sensor **107** is coupled to the wireless transmitter by wire, which is embedded in the outer surface of the pump body, the wireless transmitter transmits the pressure value to the controller by wireless signal.

(37) <Liquid Discharge Valve Assembly **103**>

(38) As shown in FIG. 3, the liquid discharge valve assembly **103** includes a liquid discharge valve base **1031**, a liquid discharge valve core **1032** slidably coupled to the liquid discharge valve base **1031**, a liquid discharge valve core stop member **1033** at the upper side of the liquid discharge valve base **1031** and a second reset spring **1034**. The liquid discharge valve core **1032** slides up and down along the liquid discharge seat under the hydraulic force, the second reset spring **1034** is used to provide elastic force downward to the liquid discharge valve core **1032**, and to block the opening of the liquid suction valve base **1021**. The liquid discharge valve core stop member **1033** plays a role of installing the second reset spring **1034** and also plays a role of blocking the liquid discharge chamber, the liquid discharge valve core stop member **1033** is axially sealed, and the axial limit is performed at the same time.

(39) In order to ensure the reliable connection between the liquid discharge valve core stop member **1033**, the upper side of the pump head body **101** is provided with a liquid discharge chamber blocking cover **105** for blocking the upper opening of the chamber, and the liquid discharge chamber blocking cover **105** presses against the liquid discharge valve core stop member **1033** and is coupled to the pump head body **101** by screws.

(40) Specifically, as shown in FIG. 7, the liquid discharge valve base **1031** is provided with a third matching surface **1031-1**. As shown in FIG. 8, the liquid discharge valve core **1032** is provided with a fourth matching surface **1032-1** that mates with the third matching surface **1031-1**. The third matching surface **1031-1** and the fourth matching surface **1032-1** are formed as conical surface; the liquid discharge valve core **1032** and the liquid discharge valve base **1031** are matched by conical surfaces, so that the sealing performance between them is better; meanwhile, the compensation can be realized automatically after being worn, and the service life is prolonged.

(41) Specifically, as shown in FIG. 3, the liquid discharge valve core stop member 1-33 is arranged between the liquid discharge valve base 1031 and the upper end of the pump head body 101. As shown in FIG. 9, the liquid discharge valve core stop member 1033 includes a spring holder part 1033-1 at the lower side, a blocking part 1033-2 at the upper side and a connecting part 1033-3 between them.

(42) The spring holder part 1033-1 includes a supporting plate extending horizontally and a supporting arm extending downward along the supporting plate, the supporting arm abuts against the upper end of the liquid discharge valve base 1031, and a convex or circular snap ring for positioning the second reset spring 1034 is formed on the supporting plate.

(43) The outer wall surface of the blocking part 1033-2 has a stepped surface, the cylindrical surface of the blocking part 1033-2 at the lower side of the stepped surface is hermetically coupled to the inner wall of the pump head body 101, the cylindrical surface of the blocking part 1033-2 located on the upper side of the stepped surface is pressed against the upper end of the pump head body 101. The structure of stepped surface of the blocking part 1033-2 can avoid the problem that the sealing performance of the pump head body 101 is not easy to be ensured since there is a gap between the liquid discharge valve core stop member 1033 and the liquid discharge chamber blocking cover 105 in the structure without stepped surface. On the other hand, the blocking part 1033-2 without a stepped surface cannot realize axial positioning, when the axial dimension of the liquid discharge valve core stop member 1033 is lower than the distance between the end of the pump head body 1-1 and the upper surface of the liquid discharge valve base 1031, it will cause the liquid discharge valve core stop member 1033 to shift up and down; axial positioning is carried out by the stepped surface of the blocking part 1033-2, which fixes the position of the liquid discharge valve core stop member 1033 relative to the pump head body 101 and reduces the machining accuracy of the liquid discharge valve core stop member 1033.

(44) Specifically, the upper end of the liquid discharge valve core 1032 is formed with a cylindrical protrusion, and one end of the reset spring is fitted over the cylindrical protrusion.

(45) In order to detect the linear displacement of the liquid discharge valve core 1032, as shown in FIG. 3, the liquid discharge chamber blocking cover 105 and the liquid discharge valve core stop member 1033 are provided with through holes communicating with each other from top to bottom, and the liquid discharge valve core 1032 is provided with threaded holes; a magnetostrictive sensor 108 is installed on the liquid discharge chamber blocking cover 105, and the telescopic rod of the magnetostrictive sensor 108 is coupled to a transition connecting rod, and the lower end of the transition connecting rod is coupled to the threaded hole of the liquid discharge valve core 1032 in a threaded manner. In this way, the fixed connection between the telescopic rod of the magnetostrictive sensor 108 and the liquid discharge valve core 1032 is realized, and the magnetostrictive sensor can calculate the operating frequency according to the displacement stroke variation curve of the liquid discharge valve core 1032, when the displacement stroke and the frequency are greatly deviated from the normal state, it can be determined that the group of the liquid suction and discharge valve core have failed, and an alarm will remind the operator to disassemble and check.

(46) <Separation Sleeve 104>

(47) As shown in FIG. 3, the separation sleeve 104 is formed as a cylinder sleeve, the cylinder sleeve has two coaxially through holes in the radial direction

(48) As shown in FIG. 3, the pump head body 101 is formed with a first through hole 1015 and a second through hole 1016 which are coaxial in the radial direction in the region of the separation sleeve mounting portion 1013, the first through hole 1015 is used to communicate with the hydraulic conversion assembly; the second through hole 1016 is provided with an anti-rotation member anti-rotation member 113 for preventing the separation sleeve 104 from rotating and a blocking member blocking member 106 for fixing the anti-rotation member anti-rotation member 113.

(49) In order to detect a vibration signal of the pump head body **101**, the pump head body **101** further provided with a vibration sensor **109**. Specifically, as shown in FIG. 3, in the middle of the blocking member blocking member **106**, there is a through hole, the anti-rotation member anti-rotation member **113** is provided with a threaded mounting hole, the vibration sensor **109** is screwed to the threaded mounting hole of the anti-rotation member anti-rotation member **113**. The vibration sensor **109** detect the vibration signal of the pump head body **101** in real time every time the liquid discharge valve core **1032** moves, according to the vibration signal, the acceleration spectrum curve is calculated. When the vibration spectrum is greatly deviated from the normal state, it can be determined that the group of the liquid discharge valve core **1032** have failed, and an alarm is given to remind the operator to disassemble and check.

(50) In the embodiments of the present disclosure, the pressure sensor **107**, the magnetostrictive sensor **108** and the vibration sensor **109** are integrated in the pump head assembly, so that the impact force, displacement stroke and vibration of the pump station are monitored, which provides a new monitoring scheme and analysis means for a running status of the pump station.

(51) <Hydraulic Conversion Assembly **200**>

(52) As shown in FIG. 1, the hydraulic conversion assembly **200** includes a hydraulic conversion housing **201** and a plunger **202** slidably connected in the hydraulic conversion housing **201**.

(53) The hydraulic conversion housing **201** is fitted to the outer side of the first through hole **1015** of the pump head body **101**; one end of the plunger **202** is coupled to the crankcase assembly **300**, the other end is inserted into the first through hole **1015**.

(54) A middle box **203** is arranged outside the hydraulic conversion assembly **200**. The crankcase assembly **300** is coupled to the pump head body **101** by the middle box **203**, the part of the plunger **202** coupled to the crankcase assembly **300** is located inside the middle box **203** to prevent this part from being exposed outside.

(55) <Crankcase Assembly **300**>

(56) As shown in FIG. 1, the crankcase assembly **300** includes: a crankcase body **301** and at least one gear pair and a crank-slider mechanism located in the crankcase body **301**; the input gear shaft **302** of the gear pair is coupled to the output shaft of the main drive motor; the crankshaft **304** of the crank-slider mechanism is coupled to the output gear **303** of the gear pair, and the slider **305** of the slider-crank mechanism is used to coupled to the plunger **202**.

(57) <Connection Mode of Crankcase Assembly **300** and Hydraulic Conversion Assembly **200**>

(58) As shown in FIG. 4, the end of the slider **305** of the crankcase assembly **300** is provided with a mounting hole, a reinforcing sleeve **306** is embedded in the mounting hole; the plunger **202** is threaded to the reinforcing sleeve **306**. Since the plunger **202** generally is made of ceramic material, while the slider **305** is made of stainless steel material, a reinforcing sleeve **306** is arranged at the connecting end of the slider **305** and the plunger **202** to solve the problem that the plunger **202** and the slider **305** are directly coupled to the plunger **202** through threads, which leads to thread tripping and fracture, improve the coupling strength between the plunger **202** and the slider **305** is beneficial to the replacement and maintenance of the plunger **202**. Specifically, the reinforcing sleeve is a steel screw sleeve with threads arranged inside and outside the reinforcing sleeve.

(59) Since the plunger **202** is usually made of ceramic material while the slider **305** is made of stainless steel material, the direct contact between them is prone to the problem of both losses, a protective sheet **400** is arranged between the slider **305** and the end face of the plunger **202**. The protective sheet **400** is made of a material softer than ceramic material and stainless steel material, such as rubber, and is squeezed between them to prevent the plunger **202** from being damaged and damaging the slider **305**.

(60) <Working Process of Plunger Pump A>

(61) When the plunger pump A is working, after the rotary motion input by the main drive motor drives the input gear shaft **302** and the output gear **303** on the crankshaft **304** to make a first-stage

deceleration motion, meanwhile, the crankshaft **304** drives the connecting rod to rotate, which is converted into the reciprocating linear motion of the slider **305** and the plunger **202**, resulting in the change of volume of the chamber in the pump head assembly **100**, when the slider **305** is at the farthest end, the plunger **202** causes the volume of the chamber to increase to form negative pressure, and the liquid suction valve core **1022** is opened and the liquid discharge valve core **1032** is closed, which completes the suction; when the slider is at the nearest end, the plunger **202** causes the volume of the chamber to decrease, thus compressing the sucked liquid to form high-pressure liquid, the liquid suction valve core **1022** is closed, and the liquid discharge valve core **1032** is opened to discharge the high-pressure liquid. This process completes the liquid discharge, the control of high-pressure hydraulic pressure is realized by unloading valves **110** and safety valves **111** installed on both sides of the pump head, the above two processes are the interactive and dynamic completion of five groups of plungers **202**. The good sealing performance in the hydraulic conversion system prevents the liquid from leaking out, and the conical sealing between the liquid suction valve core **1022** and the liquid suction valve base **1021**, and the conical sealing between the liquid discharge valve core **1032** and the liquid discharge valve base **1031** in the pump head assembly **100** jointly ensure the high volumetric efficiency of the five-plunger pump A, and make the discharged liquid flow reach more than 1200 L/min, thus meeting the actual liquid supply requirements of the hydraulic system in the super-high mining face.

(62) <Process of the Installation and Disassemble of Plunger Pump A>

(63) The pump head assembly of the emulsion plunger pump in the present disclosure is assembled in the following steps.

(64) At step **1**: five groups of the liquid suction valve assemblies **102** are assembled for later use, and the liquid suction valve base **1021** and the liquid suction valve core **1022** are grinded and undergo a sealing test before being assembled.

(65) At step **2**: two O-shaped retaining rings and one O-shaped ring are inserted into a sealing ring groove of the liquid discharge valve base **1031** in such a way that one O-shaped retaining ring is mounted on each side of the O-shaped ring, five groups of the liquid discharge valve assemblies **103** are assembled for later use, and lubrication grease is applied to a surface of the O-shaped ring for later use after the assembly is completed.

(66) At step **3**: a combined sealing ring is inserted into the sealing ring groove of the anti-rotation member anti-rotation member **113**; the above components need to be assembled in five groups; and lubrication grease is applied to surfaces of the combined sealing ring for later use after the assembly is completed.

(67) At step **4**: the combined sealing ring is inserted in a sealing ring groove of the liquid discharge valve core stop member **1033**; the above components need to be assembled in five groups; and lubrication grease is applied to surfaces of the combined sealing ring for later use after the assembly is completed.

(68) At step **5**: the liquid suction valve assembly **102** completed in step **1** is put into the liquid suction valve assembly **102** through the upper opening of the pump head body **101** and is tapped gently with a copper bar to make it closely fitted with the pump head body **101**.

(69) At step **6**: the separation sleeve **104** is placed on the liquid suction valve assembly **102** installed in step **5** through the upper opening of the pump head body **101**; the anti-rotation member anti-rotation member **113** and the blocking member blocking member **106** are mounted at the second through hole **1016**; the anti-rotation member anti-rotation member **113** is inserted into the small hole of the separation sleeve **104**; the separation sleeve **104** is positioned circumferentially and pressed with the blocking member blocking member **106**.

(70) As step **7**: the liquid discharge valve base **1031** completed in step **2** is mounted onto the upper end of the separation sleeve **104** through the assembly completed in step **6**, and is tapped gently with a copper bar to make it closely fitted with the separation sleeve **104**.

(71) At step **8**: the liquid discharge valve core **1032** and the second reset spring **1034** are mounted

inside the assembly completed in step 7.

(72) At step 9: the liquid discharge valve core stop member **1033** completed in the step 4 is mounted inside the assembly completed in step 8 and is tapped gently with a copper bar to make it closely fitted with the liquid discharge valve base **1031**, and then is pressed by the blocking cover **105** to complete the installation of the pump head assembly **100**.

(73) When the liquid discharge valve base needs to be disassembled, the following steps should be taken.

(74) At step 1: the liquid discharge chamber blocking cover **105** is disassembled.

(75) At step 3: the liquid discharge valve assembly **103** is sequentially disassembled from the upper side of the pump head body **101**.

(76) At step 3: the blocking member blocking member **106** and the anti-rotation member anti-rotation member **113** are sequentially disassembled.

(77) At step 4: the plunger **202** is disassembled from the side of the slider **305** along the first through hole **1015** and the second through hole **1016**.

(78) At step 5: the separation sleeve **104** and the liquid suction valve assembly **102** are sequentially disassembled from the upper side of the pump head body **101**.

(79) So far, the disassembly of each valve group inside the pump head body **101** has been completed. In the present disclosure, the disassembly and assembly of each part can be completed from one side of the pump head body **101**, which is convenient for maintenance.

(80) As shown in FIG. 10, a specific embodiment of a five-plunger emulsion pump station according to the present disclosure includes a base **1**, a plunger pump A and a main drive motor **2** located on the base **1**. The main drive motor **2** is used to drive the plunger pump A, and the plunger pump A adopts the plunger pump A of the above specific embodiment, which is not described here.

(81) The emulsion pump station also includes a lubrication system, which is used to lubricate the components in the crankcase assembly **300** of the plunger pump A, such as crankshaft, gear, connecting rod, bearing, slider, etc. In order to fully cool the lubrication oil during operation, the lubrication system includes a lubrication oil supply device and a cooling device.

(82) The lubrication oil supply device includes a lubrication oil pump **5**, a lubrication oil pump drive motor **4** and a lubrication oil tank (not shown in the figure); The cooling device includes: a cooling tank **3**, in which an inlet of the cooling tank **3** is in connection with the lubrication oil pump **5**, and an outlet of the cooling tank **3** is in connection with various lubrication points of the crankshaft, gear, connecting rod, bearing, slider and other parts in the crankcase assembly **300**; and an emulsion pipeline (not shown) in the cooling tank **3**, the emulsion pipeline being in connection with the pump head assembly **100** and configured to cool the lubrication oil in the cooling tank **3**.

(83) The pump station of the present disclosure adopts a cooling mode outside the plunger pump A, and a lubrication oil pump with lower power can be used to drive the motor **4**, which solves problems of large oil flow resistance and low cooling efficiency of the existing cooler arranged in the liquid suction box, and meanwhile, the implementation cost is low, and a small size of the motor is more conducive to installation and maintenance in a narrow space.

(84) Compared with the related art, the technical solutions of the present disclosure has the following technical effects.

(85) For the plunger pump in the present disclosure, in the chamber inside the pump head body, the liquid discharge valve base mounting portion is transitioned to the separation sleeve mounting portion through the stepped surface, and the bore diameter of the liquid discharge valve base mounting portion is larger than that of the separation sleeve mounting portion, and the bore diameter of the separation sleeve mounting portion is larger than or equal to that of the liquid suction valve base mounting portion. Consequently, the liquid suction valve core assembly, the separation sleeve and the liquid discharge valve core assembly can be installed from one side of the pump head body, and compared with the related art, the assembly and disassembly is convenient with less time consumption. Meanwhile, in the present disclosure, the liquid suction chamber is

directly formed in the pump head body, so there is no need for additional processing and assembly of any liquid suction box, and only a thinner blocking plate is needed to block the lower opening, which causes low manufacturing cost and simple and convenient assembly.

(86) Furthermore, in the plunger pump according to the present disclosure, the pressure sensor, the magnetostrictive sensor and the vibration sensor are integrally mounted in the plunger pump, so that the impact force, displacement stroke and vibration of the pump station are monitored, which provides a new monitoring scheme and analysis means for a running status of the pump station.

(87) Furthermore, in the pump station according to the present disclosure, a cooling mode outside the plunger pump is adopted, and a lubrication oil pump with lower power can be used to drive the motor, which solves problems of large oil flow resistance and low cooling efficiency of the existing cooler arranged in the liquid suction box, and meanwhile, the implementation cost is low, and a small size of the motor is more conducive to installation and maintenance in a narrow space.

(88) Certainly, the above embodiments are only examples for clear explanation, rather than limitations on the implementations. For those skilled in the art, other changes or variations can be made on the basis of the above description. It is not necessary or impossible to exhaust all the embodiments here. However, obvious changes or variations derived therefrom are still within the protection scope of the present disclosure.

Claims

1. A plunger pump, comprising: a crankcase assembly configured to be coupled to a main drive motor to transmit power; a pump head assembly configured to pump emulsion; and a hydraulic conversion assembly coupled between the crankcase assembly and the pump head assembly and configured to convert mechanical power of the crankcase assembly into a hydraulic change of the pump head assembly, wherein the pump head assembly comprises: a pump head body having a chamber; and a liquid suction valve assembly, a liquid discharge valve assembly and a separation sleeve separating the liquid suction valve assembly from the liquid discharge valve assembly in the chamber, wherein the chamber comprises a liquid suction valve base mounting portion for mounting the liquid suction valve assembly, a separation sleeve mounting portion for mounting the separation sleeve, a liquid discharge valve base mounting portion for mounting the liquid discharge valve assembly, and a liquid suction chamber at a lower side of the liquid suction valve base mounting portion, wherein the liquid suction chamber is in connection with an emulsion supply device through a pipeline; and the liquid discharge valve base mounting portion is transitioned to the separation sleeve mounting portion through a stepped surface; a bore diameter of the liquid discharge valve base mounting portion is larger than a bore diameter of the separation sleeve mounting portion; and the bore diameter of the separation sleeve mounting portion is larger than or equal to a bore diameter of the liquid suction valve base mounting portion; wherein the liquid discharge valve assembly comprises: a liquid discharge valve base comprising a third matching surface; a liquid discharge valve core slidably coupled to the liquid discharge valve base and comprising a fourth matching surface fitted with the third matching surface, wherein the third matching surface and the fourth matching surface are formed as conical surfaces; a liquid discharge valve core stop member between the liquid discharge valve base and an upper end face of the pump head body; and a second reset spring having a first end fitted over the liquid discharge valve core and a second end abutting against the liquid discharge valve core stop member: wherein a liquid discharge chamber blocking cover is on an upper side of the pump head body and is configured to block an upper opening of the chamber, and the liquid discharge chamber blocking cover is pressed against the liquid discharge valve core stop member and coupled to the pump head body by a screw; wherein the liquid discharge chamber blocking cover and the liquid discharge valve core stop member comprise through holes in connection with each other in a vertical direction, and the liquid discharge valve core comprises a threaded hole; and a magnetostrictive sensor is mounted on

the liquid discharge chamber blocking cover, and a telescopic rod of the magnetostrictive sensor is fixedly coupled to the liquid discharge valve core.

2. The plunger pump according to claim 1, wherein the liquid suction valve assembly comprises: a liquid suction valve base comprising a first matching surface; and a liquid suction valve core slidably coupled to the liquid suction valve base and comprising a second matching surface fitted with the first matching surface, wherein the first matching surface and the second matching surface are formed as conical surfaces.

3. The plunger pump according to claim 2, wherein the liquid suction valve assembly further comprises: a reset spring mounting seat at a lower side of the liquid suction valve base, the reset spring mounting seat being fitted over the liquid suction valve core; and a first reset spring between the liquid suction valve base and the reset spring mounting seat.

4. The plunger pump according to claim 3, wherein the liquid suction valve base comprises: an outer sleeve body, an outer wall of the outer sleeve body being hermetically coupled to the liquid suction valve base mounting portion, and an end of an inner wall of the outer sleeve body comprising the first matching surface; and an inner sleeve body coupled to the outer sleeve body through a transition arm, an inner wall of the inner sleeve body being slidably coupled to the liquid suction valve core.

5. The plunger pump according to claim 4, wherein the liquid suction valve core comprises: a valve core head comprising the second matching surface; and a valve core rod slidably coupled inside the inner sleeve body of the liquid suction valve base.

6. The plunger pump according to claim 2, wherein the liquid suction valve base comprises a pressure sensor mounting portion for mounting a pressure sensor, at a bottom of the liquid suction valve base, and the pressure sensor is configured to detect impact force during movement of the liquid suction valve core.

7. The plunger pump according to claim 1, wherein: the pump head body comprises a first through hole and a second through hole extending coaxially along a radial direction in a region of the separation sleeve mounting portion; the first through hole is in connection with the hydraulic conversion assembly; and an anti-rotation member for preventing rotation of the separation sleeve and a blocking member for fixing the anti-rotation member are disposed in the second through hole.

8. The plunger pump according to claim 7, wherein: the blocking member comprises a through hole in a middle of the blocking member; and the anti-rotation member comprises a mounting hole for mounting a vibration sensor, and the vibration sensor is configured to detect vibration of the pump head body.

9. The plunger pump according to claim 7, wherein the hydraulic conversion assembly comprises: a hydraulic conversion housing fixedly coupled to an outer side at the first through hole of the pump head body; and a plunger slidably coupled inside the hydraulic conversion housing, and having a first end coupled to the crankcase assembly and a second end inserted into the first through hole.

10. The plunger pump according to claim 9, wherein: the crankcase assembly comprises a crankcase body, and at least one gear pair and a crank-slider mechanism in the crankcase body; and an input gear of the at least one gear pair is coupled to an output shaft of the main drive motor, a crankshaft of the crank-slider mechanism is coupled to an output gear of the at least one gear pair, and a slider of the crank-slider mechanism is coupled to the plunger, wherein an end of the slider comprises a mounting hole, and a reinforcing sleeve is embedded in the mounting hole; the plunger is threadedly coupled to the reinforcing sleeve; a protective sheet is arranged between the slider and an end face of the plunger and is squeezed between the slider and the end face of the plunger.

11. A pump station, comprising: a base; and a plunger pump and a main drive motor being both on the base, wherein the main drive motor is configured to drive the plunger pump, wherein the plunger pump comprises: a crankcase assembly configured to be coupled to the main drive motor to transmit power; a pump head assembly configured to pump emulsion; and a hydraulic

conversion assembly coupled between the crankcase assembly and the pump head assembly and configured to convert mechanical power of the crankcase assembly into a hydraulic change of the pump head assembly, wherein the pump head assembly comprises: a pump head body having a chamber; and a liquid suction valve assembly, a liquid discharge valve assembly and a separation sleeve separating the liquid suction valve assembly from the liquid discharge valve assembly in the chamber, wherein the chamber comprises a liquid suction valve base mounting portion for mounting the liquid suction valve assembly, a separation sleeve mounting portion for mounting the separation sleeve, a liquid discharge valve base mounting portion for mounting the liquid discharge valve assembly, and a liquid suction chamber at a lower side of the liquid suction valve base mounting portion, wherein the liquid suction chamber is in connection with an emulsion supply device through a pipeline; and the liquid discharge valve base mounting portion is transitioned to the separation sleeve mounting portion through a stepped surface; a bore diameter of the liquid discharge valve base mounting portion is larger than a bore diameter of the separation sleeve mounting portion; and the bore diameter of the separation sleeve mounting portion is larger than or equal to a bore diameter of the liquid suction valve base mounting portion; wherein the liquid discharge valve assembly comprises: a liquid discharge valve base comprising a third matching surface; a liquid discharge valve core slidably coupled to the liquid discharge valve base and comprising a fourth matching surface fitted with the third matching surface, wherein the third matching surface and the fourth matching surface are formed as conical surfaces; a liquid discharge valve core stop member between the liquid discharge valve base and an upper end face of the pump head body; and a second reset spring having a first end fitted over the liquid discharge valve core and a second end abutting against the liquid discharge valve core stop member; wherein a liquid discharge chamber blocking cover is on an upper side of the pump head body and is configured to block an upper opening of the chamber, and the liquid discharge chamber blocking cover is pressed against the liquid discharge valve core stop member and coupled to the pump head body by a screw; wherein the liquid discharge chamber blocking cover and the liquid discharge valve core stop member comprise through holes in connection with each other in a vertical direction, and the liquid discharge valve core comprises a threaded hole; and a magnetostrictive sensor is mounted on the liquid discharge chamber blocking cover, and a telescopic rod of the magnetostrictive sensor is fixedly coupled to the liquid discharge valve core.

12. The pump station according to claim 11, wherein the liquid suction valve assembly comprises: a liquid suction valve base comprising a first matching surface; and a liquid suction valve core slidably coupled to the liquid suction valve base and comprising a second matching surface fitted with the first matching surface, wherein the first matching surface and the second matching surface are formed as conical surfaces.

13. The pump station according to claim 12, wherein the liquid suction valve assembly further comprises: a reset spring mounting seat at a lower side of the liquid suction valve base, the reset spring mounting seat being fitted over the liquid suction valve core; and a first reset spring between the liquid suction valve base and the reset spring mounting seat.

14. The pump station according to claim 13, wherein the liquid suction valve base comprises: an outer sleeve body, an outer wall of the outer sleeve body being hermetically coupled to the liquid suction valve base mounting portion, and an end of an inner wall of the outer sleeve body comprising the first matching surface; and an inner sleeve body coupled to the outer sleeve body through a transition arm, an inner wall of the inner sleeve body being slidably coupled to the liquid suction valve core.

15. A plunger pump, comprising: a crankcase assembly configured to be coupled to a main drive motor to transmit power; a pump head assembly configured to pump emulsion; and a hydraulic conversion assembly coupled between the crankcase assembly and the pump head assembly and configured to convert mechanical power of the crankcase assembly into a hydraulic change of the pump head assembly, wherein the pump head assembly comprises: a pump head body having a

chamber; and a liquid suction valve assembly, a liquid discharge valve assembly and a separation sleeve separating the liquid suction valve assembly from the liquid discharge valve assembly in the chamber, wherein the chamber comprises a liquid suction valve base mounting portion for mounting the liquid suction valve assembly, a separation sleeve mounting portion for mounting the separation sleeve, a liquid discharge valve base mounting portion for mounting the liquid discharge valve assembly, and a liquid suction chamber at a lower side of the liquid suction valve base mounting portion, wherein the liquid suction chamber is in connection with an emulsion supply device through a pipeline; and the liquid discharge valve base mounting portion is transitioned to the separation sleeve mounting portion through a stepped surface; a bore diameter of the liquid discharge valve base mounting portion is larger than a bore diameter of the separation sleeve mounting portion; and the bore diameter of the separation sleeve mounting portion is larger than or equal to a bore diameter of the liquid suction valve base mounting portion; wherein: the pump head body comprises a first through hole and a second through hole extending coaxially along a radial direction in a region of the separation sleeve mounting portion; the first through hole is in connection with the hydraulic conversion assembly; and an anti-rotation member for preventing rotation of the separation sleeve and a blocking member for fixing the anti-rotation member are disposed in the second through hole.
